

LOCKHART RIVER, Australia

WMO: 941860

Lat: 12.7850S

Lon: 143.3047E

Elev: 18

StdP: 101.11

Time Zone: 10.00 (E10)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.4	16.7	14.3	10.2	18.1	15.3	10.9	19.5	9.7	25.9	9.3	25.8	0.1	280	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.9	33.1	26.0	32.1	25.8	31.4	25.5	27.1	30.5	26.7	30.0	26.4	29.6	4.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.3	25.9	21.2	28.0	25.6	20.8	27.8	85.3	30.6	83.9	30.2	82.5	29.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	8.3	7.6	DB	13.3	35.3	2.3	1.4	11.7	36.3	10.3	37.1	9.0	37.8	7.3	38.9	(4)
(5)			WB	13.2	28.0	2.2	0.5	11.6	28.3	10.3	28.6	9.1	28.8	7.5	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	27.6	27.4	26.9	26.3	25.1	23.9	23.3	23.1	24.7	26.0	27.3	27.9	(6)	
	DBStd	2.13	1.25	1.11	1.03	1.05	1.39	1.62	1.56	1.57	1.61	1.44	1.18	1.11	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	5762	544	488	524	488	469	417	413	407	442	497	518	555	(10)	
	CDD18.3	2721	286	255	266	238	211	167	154	148	192	238	268	297	(11)	
	CDH23.3	23271	2803	2437	2392	2101	1566	895	681	771	1403	2146	2794	3283	(12)	
(13)	CDH26.7	5893	942	766	652	428	181	58	17	42	199	501	869	1237	(13)	
(14)	Wind	WSAvg	3.7	2.6	2.3	2.3	3.7	4.3	3.9	4.4	4.2	4.9	4.5	4.1	3.1	(14)
(15) Precipitation	PrecAvg	2038	383	407	439	294	104	58	42	26	18	20	82	208	(15)	
	PrecMax	3298	784	783	1286	1178	485	205	136	106	87	99	521	717	(16)	
	PrecMin	1136	104	110	83	25	10	4	0	0	0	1	0	15	(17)	
	PrecStd	555	167	159	254	243	100	47	25	22	22	22	118	148	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.2	32.7	30.7	29.2	28.8	27.8	28.5	29.7	31.7	33.5	34.2	(19)	
		MCWB	25.9	26.2	26.4	25.4	24.2	24.2	22.9	23.2	23.9	24.6	26.0	25.8	(20)	
	2%	DB	32.7	32.7	31.4	29.9	28.7	28.0	27.2	27.7	29.0	30.4	31.5	32.7	(21)	
		MCWB	26.1	26.3	26.2	25.0	24.1	23.6	22.4	22.2	23.2	24.3	25.0	25.6	(22)	
	5%	DB	31.6	31.4	30.6	29.4	28.3	27.3	26.7	27.1	28.4	29.7	30.7	31.9	(23)	
		MCWB	25.8	26.0	25.9	24.8	23.9	23.2	22.2	21.8	22.7	23.9	24.5	25.4	(24)	
	10%	DB	30.7	30.5	29.8	28.9	27.8	26.7	26.2	26.6	27.9	29.1	30.2	31.2	(25)	
		MCWB	25.6	25.9	25.6	24.6	23.7	22.9	22.1	21.5	22.3	23.5	24.2	25.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.5	27.5	26.8	26.0	25.2	24.7	25.0	25.6	26.0	26.5	27.1	(27)	
		MCDB	30.9	31.5	30.6	28.9	27.8	27.3	26.3	27.1	28.2	29.9	31.1	31.2	(28)	
	2%	WB	26.8	26.9	27.0	26.2	25.2	24.6	24.1	24.0	24.9	25.2	25.9	26.6	(29)	
		MCDB	30.4	30.5	29.8	28.3	27.2	26.6	25.7	25.8	27.1	28.6	29.9	30.4	(30)	
	5%	WB	26.5	26.5	26.6	25.9	24.8	24.1	23.6	23.1	24.1	24.8	25.4	26.2	(31)	
		MCDB	29.9	30.0	29.3	27.8	26.9	26.1	25.4	25.5	27.0	28.1	29.2	29.9	(32)	
	10%	WB	26.1	26.2	26.2	25.5	24.4	23.7	23.0	22.5	23.3	24.4	25.0	25.9	(33)	
		MCDB	29.4	29.4	28.7	27.4	26.6	25.7	25.1	25.2	26.8	27.7	28.7	29.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.4	5.9	5.4	5.5	5.6	5.9	7.2	6.7	6.6	6.5	6.9	(35)	
		MCDBR	8.0	8.1	7.3	6.3	6.2	6.2	6.6	8.2	7.2	7.3	7.1	8.0	(36)	
	5% WB	MCWBR	2.7	3.1	2.9	2.7	2.9	2.8	3.1	3.8	3.0	2.8	2.4	2.5	(37)	
		MCWBR	7.0	6.9	6.4	5.0	4.5	4.6	4.0	5.1	5.4	6.3	6.8	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.405	0.399	0.397	0.400	0.399	0.393	0.382	0.389	0.400	0.412	0.414	0.412	(40)	
		taud	2.534	2.555	2.559	2.528	2.496	2.508	2.524	2.488	2.450	2.417	2.439	2.480	(41)	
	Ebn at Noon		939	937	921	886	854	843	862	883	904	914	923	930	(42)	
		Edh at Noon	111	109	105	103	101	97	97	106	116	123	122	117	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.38	5.41	4.98	4.92	4.53	4.05	4.37	5.17	6.01	6.51	6.59	6.13	(44)	
	RadStd		0.56	0.63	0.35	0.49	0.33	0.32	0.30	0.43	0.33	0.40	0.40	0.61	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	+112	+109

Nomenclature: See separate page